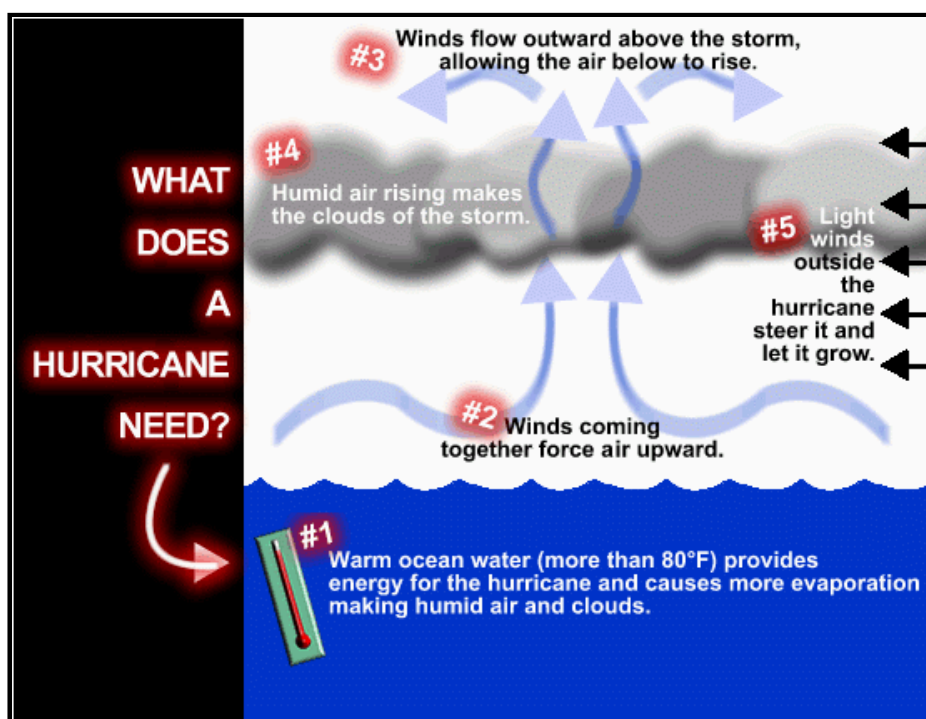


Hurricane Facts



There are six widely accepted conditions for hurricane development:

1. The first condition is that ocean waters must be above 26 degrees Celsius (79 degrees Fahrenheit). Below this threshold temperature, hurricanes will not form or will weaken rapidly once they move over water below this threshold. Ocean temperatures in the tropical East Pacific and the tropical Atlantic routinely surpass this threshold.
2. The second ingredient is distance from the equator. Without the spin of the earth and the resulting Coriolis force, hurricanes would not form. Since the Coriolis force is at a maximum at the poles and a minimum at the equator, hurricanes can not form within 5 degrees latitude of the equator. The Coriolis force generates a counterclockwise spin to low pressure in the Northern Hemisphere and a clockwise spin to low pressure in the Southern Hemisphere.
3. The third ingredient is that of a saturated lapse rate gradient near the center of rotation of the storm. A saturated lapse rate insures latent heat will be released at a maximum rate. Hurricanes are warm core storms. The heat hurricanes generate is from the condensation of water vapor as it convectively rises around the eye wall. The lapse rate must be unstable around the eyewall to insure rising parcels of air will continue to rise and condense water vapor.
4. The fourth and one of the most important ingredients is that of a low vertical wind shear, especially in the upper level of the atmosphere. Wind shear is a change in wind speed with height. Strong upper level winds destroy the storms structure by displacing the warm temperatures above the eye and limiting the vertical ascent of air parcels. Hurricanes will not form when the upper level winds are too strong.
5. The fifth ingredient is high relative humidity values from the surface to the mid levels of the atmosphere. Dry air in the mid levels of the atmosphere impedes hurricane development in two ways. First, dry air causes evaporation of liquid water. Since evaporation is a cooling process, it reduces the warm core structure of the hurricane and limits vertical development of convection. Second, dry air in the mid levels can create what is known as a trade wind inversion. This inversion is similar to sinking air in a high pressure system. The trade wind inversion produces a layer of warm temperatures and dryness in the mid levels of the atmosphere due to the sinking and adiabatic warming of the mid level air. This inhibits deep convection and produces a stable lapse rate.
6. The sixth ingredient is that of a tropical wave. Often hurricanes in the Atlantic begin as a thunderstorm complex that moves off the coast of Africa. It becomes what is known as a midtropospheric wave. If this wave encounters favorable

conditions such as stated in the first five ingredients, it will amplify and evolve into a tropical storm or hurricane. Hurricanes in the East Pacific can develop by a midtropospheric wave or by what is known as a monsoonal trough.

Additional items...

- Each year, an average of ten tropical storms develop over the Atlantic Ocean, Caribbean Sea, and Gulf of Mexico. Many of these remain over the ocean. Six of these storms become hurricanes each year. In an average 3-year period, roughly five hurricanes strike the United States coastline, killing approximately 50 to 100 people anywhere from Texas to Maine. Of these, two are typically major hurricanes (winds greater than 110 mph).
- Typical hurricanes are about 300 miles wide although they can vary considerably in size.
- The eye at a hurricane's center is a relatively calm, clear area approximately 20-40 miles across.
- The eyewall surrounding the eye is composed of dense clouds that contain the highest winds in the storm.
- The storm's outer rainbands (often with hurricane or tropical storm-force winds) are made up of dense bands of thunderstorms ranging from a few miles to tens of miles wide and 50 to 300 miles long.
- Hurricane-force winds can extend outward to about 25 miles in a small hurricane and to more than 150 miles for a large one. Tropical storm-force winds can stretch out as far as 300 miles from center of a large hurricane.
- Frequently, the right side of a hurricane is the most dangerous in terms of storm surge, winds, and tornadoes.
- A hurricane's speed and path depend on complex ocean and atmospheric interactions, including the presence or absence of other weather patterns. This complexity of the flow makes it very difficult to predict the speed and direction of a hurricane.
- Do not focus on the eye or the track-hurricanes are immense systems that can move in complex patterns that are difficult to predict. Be prepared for changes in size, intensity, speed and direction.

